

Week7

Task1:

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
#include <stdio.h>
#include <stdlib.h>

/* Global variable */
int global_var = 10;

/* Static global variable */
static int static_global = 20;

/* Function - Code/Text segment */
void display()
{
    printf("This is a function\n");
}

int main()
{
    /* Stack variable */
    int stack_var = 30;

    /* Static local variable */
    static int static_local = 40;

    /* Heap variable */
    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 50;

    printf("----- MEMORY ADDRESSES ----- \n");

    /* Code segment */
    printf("Code address          : %p\n", (void *)display);

    /* Global segment */
    printf("Global variable          : %p\n", (void *)&global_var);

    /* Static global */
    printf("Static global              : %p\n", (void *)&static_global);

    /* Static local */
    printf("Static local               : %p\n", (void *)&static_local);
}
```

```

/* Static global */
printf("Static global      : %p\n", (void *)&static_global);

/* Static local */
printf("Static local      : %p\n", (void *)&static_local);

/* Heap */
printf("Heap variable     : %p\n", (void *)heap_var);

/* Stack */
printf("Stack variable    : %p\n", (void *)&stack_var);

free(heap_var);

return 0;
}

```

Output:

```

root@DESKTOP-VMN932K:~# ./n1
----- MEMORY ADDRESSES -----
Code address      : 0x5cc523a831c9
Global variable   : 0x5cc523a86010
Static global     : 0x5cc523a86014
Static local      : 0x5cc523a86018
Heap variable     : 0x5cc561de22a0
Stack variable    : 0x7ffdd229b1cc
root@DESKTOP-VMN932K:~# █

```

Task2:

Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```

#include <stdio.h>
#include <unistd.h>

int main()
{
    FILE *file;
    char filename[50];
    char line[256];

    /* Get process ID */
    pid_t pid = getpid();

    printf("Process ID: %d\n", pid);
    /* Create /proc/PID/maps path */
    sprintf(filename, "/proc/%d/maps", pid);
    /* Open memory map */
    file = fopen(filename, "r");
    if (file == NULL)
    {
        perror("Unable to open /proc/PID/maps");
        return 1;
    }
    printf("\n----- PROCESS MEMORY MAP ----- \n\n");
    /* Read and display each line */
    while (fgets(line, sizeof(line), file))
    {
        printf("%s", line);
    }
    fclose(file);
    return 0;
}

```

Output:

```

root@DESKTOP-VMN932K:~# ./n2
Process ID: 3149

----- PROCESS MEMORY MAP -----

5742b1c11000-5742b1c12000 r--p 00000000 08:30 64979 /root/n2
5742b1c12000-5742b1c13000 r-xp 00001000 08:30 64979 /root/n2
5742b1c13000-5742b1c14000 r--p 00002000 08:30 64979 /root/n2
5742b1c14000-5742b1c15000 r--p 00002000 08:30 64979 /root/n2
5742b1c15000-5742b1c16000 rw-p 00003000 08:30 64979 /root/n2
5742b2b21000-5742b2b24000 rw-p 00000000 00:00 0 [heap]
7216f2e00000-7216f2e28000 r--p 00000000 08:30 44659 /usr/lib/x86_64-linux-gnu/libc.so.6
7216f2e28000-7216f2fb1000 r-xp 00028000 08:30 44659 /usr/lib/x86_64-linux-gnu/libc.so.6
7216f2fb1000-7216f3000000 r--p 001b1000 08:30 44641 /usr/lib/x86_64-linux-gnu/libc.so.6
7216f3000000-7216f3004000 r--p 001ff000 08:30 44659 /usr/lib/x86_64-linux-gnu/libc.so.6
7216f3004000-7216f3006000 rw-p 00203000 08:30 44659 /usr/lib/x86_64-linux-gnu/libc.so.6
7216f3006000-7216f3013000 rw-p 00000000 00:00 0
7216f30e9000-7216f30ec000 rw-p 00000000 00:00 0
7216f30f1000-7216f30f3000 rw-p 00000000 00:00 0
7216f30f3000-7216f30f7000 r--p 00000000 00:00 0 [vvar]
7216f30f7000-7216f30f9000 r--p 00000000 00:00 0 [vvar_vclock]
7216f30f9000-7216f30fb000 r-xp 00000000 00:00 0 [vdso]
7216f30fb000-7216f30fc000 r--p 00000000 08:30 44641 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7216f30fc000-7216f3127000 r-xp 00001000 08:30 44641 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7216f3127000-7216f3131000 r--p 0002c000 08:30 44641 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7216f3131000-7216f3133000 r--p 00036000 08:30 44641 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7216f3133000-7216f3135000 rw-p 00038000 08:30 44641 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffd09bac000-7ffd09bcd000 rw-p 00000000 00:00 0 [stack]
root@DESKTOP-VMN932K:~#

```